

A Hierarchical Bayesian Skill-Library Partial-Order Model

HPOP — formal model specification

We formalize a hierarchical Bayesian model of agent traces:

Actions $\longrightarrow$ **Local partial orders (skills)** $\longrightarrow$ **Global partial order**,

in which skills are reusable, are drawn from a shared *library*, and *creating a new skill is penalized*. The model is the hierarchical, nonparametric extension of BPOP: a Dirichlet-process mixture whose atoms are local partial orders, ordered at the top level by a global partial order. Both ordering likelihoods use the tractable robust frontier-softmax, avoiding #P-hard linear-extension counting.

1 Notation

Let $\mathcal{A} = \{1, \dots, M\}$ be the canonical action (CPA) alphabet. A corpus is a set of traces; trace n is an observed action sequence $y^{(n)} = (y_1^{(n)}, \dots, y_{T_n}^{(n)})$ with $y_t^{(n)} \in \mathcal{A}$, obtained after physical normalization and CPA abstraction (PDAF Steps 1a–1b). Latent structure per trace: a segmentation of $y^{(n)}$ into contiguous *skill instances*, a *skill-type* label per instance, a local partial order per skill type, and a global partial order over the instances.

2 Partial orders and their latent representation

Definition 1 (Strict partial order). *A strict partial order (poset) on a finite set $\mathcal{X}$ is a pair $h = (\mathcal{X}, \succ_h)$ with $\succ_h$ irreflexive and transitive. $a \parallel_h b$ (incomparable) iff neither $a \succ_h b$ nor $b \succ_h a$. A linear extension is a total order on $\mathcal{X}$ consistent with $\succ_h$; $\mathcal{L}(h)$ is the set of all linear extensions.*

Definition 2 (Latent realizer / dimension representation). *For $\mathcal{X}$ with $|\mathcal{X}| = m$ and a latent matrix $U \in \mathbb{R}^{m \times K}$, define the poset $h(U)$ by*

$$a \succ_{h(U)} b \iff U_{a,k} > U_{b,k} \quad \forall k \in \{1, \dots, K\}. \quad (1)$$

Thus $h(U)$ is the intersection of K total orders (realizers); $\dim(h(U)) \leq K$ (Dushnik–Miller). Incomparability arises when two rows fail to dominate across all K coordinates, encoding concurrency.

Prior on a poset. Following Nicholls et al., place

$$K \sim \text{tPoisson}(\lambda), \quad U_{:,k} \stackrel{iid}{\sim} \mathcal{N}(\mathbf{0}, \Sigma_\rho), \quad k = 1, \dots, K, \quad (2)$$

where Σ_ρ has unit variance and correlation ρ ; the dimension K is itself inferred. Write $p_0(h) = \int \mathbf{1}[h(U) = h] dU dK$ for the induced prior over posets.

3 The frontier-softmax likelihood

Given a poset h on $\mathcal{X}$, an agent emits its elements as a stochastic linear extension by repeatedly choosing from the *frontier*.

Definition 3 (Frontier). *For a prefix that has emitted the set $S \subseteq \mathcal{X}$, the frontier is $F(S; h) = \{a \in \mathcal{X} \setminus S : \nexists b \in \mathcal{X} \setminus S, b \succ_h a\}$, the minimal not-yet-emitted elements. Elements of $F(S; h)$ are mutually unordered given S (concurrent).*

Definition 4 (Robust frontier-softmax). *Let $w = (w_1, \dots, w_m)$ be an emitted order, $S_t = \{w_1, \dots, w_{t-1}\}$, $F_t = F(S_t; h)$, $R_t = \mathcal{X} \setminus S_t$, and the successor utility $Q(a; h) = \log(1 + |\{b : a \succ_h b\}|)$. With inverse temperature $\beta \geq 0$ and noise $\epsilon \in [0, 1)$,*

$$p(w \mid h, \beta, \epsilon) = \prod_{t=1}^m \left[(1 - \epsilon) \frac{\mathbf{1}[w_t \in F_t] e^{\beta Q(w_t; h)}}{\sum_{a \in F_t} e^{\beta Q(a; h)}} + \epsilon \frac{1}{|R_t|} \right]. \quad (3)$$

If $w_t \notin F_t$ (an order violation) the rational term vanishes and only the ϵ noise survives, so violating traces retain positive probability. Evaluating (??) costs $O(m|\mathcal{X}|)$ via Kahn’s algorithm, avoiding the #P-complete enumeration of $\mathcal{L}(h)$.

4 Skills, the library, and the base measure

A *skill* is a local partial order together with its action support: $\theta = (\mathcal{A}_\theta, U_\theta)$ with $\mathcal{A}_\theta \subseteq \mathcal{A}$ and local poset $P_\theta = h(U_\theta)$ on $\mathcal{A}_\theta$. The base measure G_0 over skills draws a support and a poset via (??).

The shared library is nonparametric:

$$G = \sum_{r \geq 1} \pi_r \delta_{\theta_r}, \quad \pi \sim \text{GEM}(\alpha), \quad \theta_r \stackrel{iid}{\sim} G_0, \quad \text{i.e. } G \sim \text{DP}(\alpha, G_0). \quad (4)$$

Each skill instance’s type is an i.i.d. draw from G ; integrating out π yields the Chinese restaurant process (CRP) predictive used below. (Corpus-level sharing with per-trace reuse is the hierarchical DP; we state the single shared-library CRP, which suffices for the penalty.)

5 Generative model

Let $z_{n,i} \in \{1, 2, \dots\}$ index the skill type of instance i in trace n , and let $(n, i) \mapsto g(n, i)$ enumerate all instances in exchangeable order with running type counts n_r .

1. **Library (lazy).** Maintain atoms $\theta_r = (\mathcal{A}_r, U_r)$, drawing a fresh $\theta_{r^*} \sim G_0$ the first time a new type is opened.
2. **Skill-type assignment (CRP).** For the $N+1$ -st instance overall,

$$P(z = r \mid z_{1:N}) = \frac{n_r}{N + \alpha}, \quad P(z = \text{new}) = \frac{\alpha}{N + \alpha}. \quad (5)$$

3. **Global poset.** For trace n with T_n instances draw $V_n \in \mathbb{R}^{T_n \times K_n^g}$ via (??) (params λ_g), giving $Q_n = h(V_n)$.

4. **Global linear extension.** Emit the instance order $\sigma_n \in \mathcal{L}(Q_n)$ from the global frontier-softmax $p(\sigma_n \mid Q_n, \beta_g, \epsilon_g)$ (??).
5. **Local linear extensions.** For each instance i of type $r = z_{n,i}$, emit its action block $y_{[i]}^{(n)}$ (an ordering of the nodes of P_r) from the local frontier-softmax $p(y_{[i]}^{(n)} \mid P_r, \beta_\ell, \epsilon_\ell)$ (??).

The observed trace $y^{(n)}$ is the concatenation of the blocks $\{y_{[i]}^{(n)}\}$ in the order σ_n : a *nested* linear extension (a linear extension of the global poset whose elements are themselves linear extensions of local posets).

6 Joint density and the new-skill penalty

With $\Theta = \{\theta_r\}$ the realized library and $\Phi = (\alpha, \lambda, \lambda_g, \beta_\ell, \beta_g, \epsilon)$,

$$p(\{y^{(n)}, \sigma_n, V_n, z_{n,\cdot}\}_n, \Theta \mid \Phi) = \underbrace{p(z_{\cdot} \mid \alpha)}_{\text{types}} \prod_r \underbrace{G_0(\theta_r)}_{\text{skill params}} \prod_n \left[\underbrace{p_0(V_n)}_{\text{global prior}} \underbrace{p(\sigma_n \mid h(V_n), \beta_g, \epsilon_g)}_{\text{global LE}} \right. \\ \left. \times \prod_{i=1}^{T_n} \underbrace{p(y_{[i]}^{(n)} \mid P_{z_{n,i}}, \beta_\ell, \epsilon_\ell)}_{\text{local LE}} \right]. \quad (6)$$

The first two factors are the cost of the *library*; the bracket is the cost of *ordering* skills and actions. We now isolate the penalty.

Proposition 1 (New-skill penalty via the EPPF). *Let $z_{1:N}$ have K distinct types with sizes $n_1, \dots, n_K$. Under the CRP (??), the type prior is the Ewens exchangeable partition probability function*

$$p(z_{1:N} \mid \alpha) = \alpha^K \frac{\Gamma(\alpha)}{\Gamma(N + \alpha)} \prod_{r=1}^K \Gamma(n_r), \quad \log p(z_{1:N} \mid \alpha) = K \log \alpha + \sum_r \log \Gamma(n_r) + \log \frac{\Gamma(\alpha)}{\Gamma(N + \alpha)}. \quad (7)$$

Hence the number of distinct skills K enters the log-joint linearly with coefficient $\log \alpha$: each new skill incurs a fixed prior penalty of $-\log \alpha$ (for $\alpha < 1$, a positive cost), while reusing a type of size n_r contributes the cheaper $\log \Gamma(n_r + 1) - \log \Gamma(n_r) = \log n_r$. Opening a new type additionally pays the marginal (Bayesian-Occam) cost

$$-\log \int p(y_{[i]} \mid h(U)) dG_0(U) \quad (8)$$

in place of reusing an existing P_r , so a new skill is favored only when it explains its block substantially better than any library skill.

Pitman–Yor generalization. Replacing the CRP by a Pitman–Yor process $\text{PY}(\alpha, d)$, $d \in [0, 1)$, gives

$$P(z = \text{new}) = \frac{\alpha + dK}{N + \alpha}, \quad P(z = r) = \frac{n_r - d}{N + \alpha}, \quad (9)$$

a per-new-skill penalty $-\log(\alpha + dK)$ that grows with K , yielding power-law library sizes when a heavier tail of rare skills is expected. The single knob α (with optional d) controls the reuse–novelty trade-off: smaller $\alpha \Rightarrow$ stronger reuse and fewer skills.

7 Posterior and inference

The posterior over (segmentation, z , $\{U_r\}$, $\{V_n\}$, Φ) given $\{y^{(n)}\}$ is proportional to (??). We target it with MCMC:

- *Type reassignment* (collapsed Gibbs, DP-mixture form): for instance (n, i) , $P(z_{n,i} = r \mid \dots) \propto (n_r^{-(n,i)} \text{ or } \alpha) \cdot p(y_{[i]}^{(n)} \mid P_r, \beta_\ell, \epsilon_\ell)$, with the global-LE factor enforcing consistency with σ_n ; the α branch is Proposition ??.
- *Skill split/merge* (library moves), *semi-Markov* resegmentation of $y^{(n)}$.
- *Dimension* K_r, K_n^g by reversible-jump MCMC (birth/death of realizer columns).
- *Continuous* updates of $U_r, V_n, \beta_\ell, \beta_g, \epsilon$ and a hyperprior on α by MH.

The recovered structure compiles into a frontier executor: incomparable elements run concurrently and reused skills are not re-planned, giving the inference-time speed-up.

8 Informative priors from PDAF annotation

PDAF supplies weak supervision that sharpens the priors:

- **local_orders** (typed CPA edges with confidence) define a per-pair prior $\pi_{ab} \in \{\succ, \prec, \parallel\}$ for each skill, replacing the uniform 1/3 prior over the three relations; entered as a prior factor / pseudo-observations on U_r . **INCOMPARABLE** contributes direct $\parallel$ evidence, the relation hardest to recover from linear extensions alone.
- **global_orders** (skill-level edges) prior on Q_n via V_n .
- **skill_name** and the library seed the atoms $\{\theta_r\}$ and bias the CRP labels z ; the annotator’s **skill_is_new** flag is the empirical counterpart of the CRP “new table” event (??) and is used as a proposal bias for opening new skills.